

An asymptotic cover law for sparse nearly linear intersecting hypergraphs

Research continuation, 7 October 2026. We give a cover bound depending on the edge/rank ratio, without a partite assumption, and describe its equality regime. The proof uses a cited published colouring theorem. There is no mathematical priority claim or claim of full formalization.

1. Statements and conventions

Let H be a finite **simple** intersecting r -uniform hypergraph, $r \geq 2$. Vertices are restricted to the union of its edges. Write $m = |E(H)|$, $t = \tau(H)$, and

$$I(H) = \sum_{\{A,B\} \subset E(H)} (|A \cap B| - 1). \quad (1)$$

Pairs in this sum are unordered and distinct. Its summands are nonnegative. For an empty family set $t = I(H) = 0$. An intersecting family is linear exactly when $I(H) = 0$. No partite assumption is made.

Define the continuous concave function

$$h(x) = \min_{k \geq 1} \left\{ \frac{k-1}{k+1} + \frac{x}{k(k+1)} \right\}, \quad x \geq 0. \quad (2)$$

On $[k-1, k]$ its displayed minimum is attained by k . In particular $h(x) = x/2$ for $0 \leq x \leq 1$, and $h(a) = a/(a+1)$ for positive integers a . It is the piecewise linear interpolation of these values, and

$$h(x) \leq \frac{x}{1+x}. \quad (3)$$

Theorem 1 (finite envelope inequality). Fix $C \geq 1$, put

$$K = \max\{2, \lceil C \rceil\}, \quad L = K(K+1), \quad D = L-1, \quad G_C = \frac{(D-1)(D-2)}{2}.$$

For every $\varepsilon > 0$ there is a finite constant $B_{C,\varepsilon}$, independent of H, r , such that, if $m \leq Cr$, then

$$t \leq rh(m/r) + G_C \frac{I(H)}{r} + \varepsilon r + B_{C,\varepsilon}. \quad (4)$$

The constant is defined through the threshold in Input K below. No effective growth rate for that threshold is asserted.

Corollary 2 (the sparse near-linear regime). Suppose $r_j \rightarrow \infty$, $m_j/r_j \rightarrow c < \infty$, and $I(H_j) = o(r_j^2)$. Then

$$\limsup_j \frac{\tau(H_j)}{r_j} \leq h(c) \leq \frac{c}{1+c}. \quad (5)$$

More generally the harmonic bound holds along a sequence with bounded m_j/r_j , without requiring convergence of that ratio. Consequently, if $\tau(H_j)/r_j \rightarrow 1$ and $I(H_j) = o(r_j^2)$, then $m_j/r_j \rightarrow \infty$.

The last assertion applies both to partite Ryser equality families $\tau \geq r - 1$ and to nearly linear families in the unrestricted Erdős–Lovász problem $\tau = r$. It concerns a restricted class; it does not contradict the unrestricted linear-in- r upper bounds for the Erdős–Lovász problem, or decide the analogous general partite conjecture.

Theorem 3 (integer equality rigidity). In the setting of Corollary 2, suppose $c = a$ is a positive integer and $\tau(H_j)/r_j \rightarrow a/(a + 1)$. Let n_j be the number of active vertices and $d_j(v)$ their degrees. Then

$$\frac{n_j}{r_j^2} \rightarrow \frac{a}{a + 1}, \quad \sum_v (d_j(v) - a - 1)^2 = o(r_j^2). \quad (6)$$

For every prime power $a + 1$ these conclusions are attained by the explicit affine-geometric families in Section 7. The interval $0 \leq c \leq 1$ in (5) is also attained.

Theorem 4 (noninteger obstruction to equality). If $c > 1$ is not an integer, no sequence in Corollary 2 can have $\tau(H_j)/r_j \rightarrow h(c)$. In fact for each such c there exists $\zeta_c > 0$ such that every sequence in that corollary satisfies

$$\limsup_j \tau(H_j)/r_j \leq h(c) - \zeta_c. \quad (7)$$

An explicit, conservative choice is given in Section 6.1. It is not claimed to be optimal. The full envelope is therefore a universal bound with attained integer endpoints, rather than a claim of a sharp frontier at every real ratio.

2. Published input and finite thresholds

We use Kahn’s small-codegree hypergraph edge-colouring theorem in the form stated in Kang–Kelly–Kühn–Methuku–Osthus [KKKMO, Theorem 3.1]. For every fixed rank D and $\eta > 0$, there is $\gamma > 0$ such that a hypergraph of rank at most D , maximum degree at most A , and maximum pair codegree at most γA has chromatic index at most $(1 + \eta)A$. Their stated theorem gives list colouring, of which ordinary colouring is a consequence.

The source explicitly allows parallel edge copies (Section 1.2): distinct edge identities may have the same vertex set. Copies count towards degrees and codegrees and cannot share a colour. This convention is needed in Section 3. Choose an integer $q_0 = q_0(D, \eta) \geq 2$ such that $D - 1 \leq \gamma(q - 1)$ for every $q \geq q_0$. Input K thus applies with the *upper bound* $A = q - 1$; it does not require the actual maximum degree to equal this bound. Empty auxiliary graphs cause no difficulty. We neither re-prove nor formalize Input K.

Peeling, pair-excess deletion and matching covers are related to the tools in Sivashankar [S, Section 4]. The earlier partite notes in this repository use replicated blocks with multiplicity $d - 3$. Here multiplicity $d - 1$ bounds the replicated degree by $q - 1$ and permits a counting argument without parts. None of the mathematical claims in OpenAI/math are used as an input.

3. A bounded-degree residual lemma

Let J be intersecting and r -uniform, with q edges, maximum vertex degree at most a fixed integer $D \geq 3$, $T = I(J)$, and n active vertices. Let x_d count its degree- d vertices. Double counting gives

$$\sum d x_d = q r, \quad \sum \binom{d}{2} x_d = \binom{q}{2} + T, \quad \sum d^2 x_d = q(q + r - 1) + 2T. \quad (8)$$

Let $g_d = (d-1)(d-2)/2$. The exact total weight is

$$F := \sum g_d x_d = \binom{q}{2} + T - qr + n. \quad (9)$$

View the q original edges as points. Every original vertex of degree $d \geq 2$ defines its incidence block of size d ; retain block identities, including repeated blocks. Their total pair excess is exactly T . Repeatedly delete a block containing a pair of codegree at least two. The pair excess decreases by at least one per deletion and never increases. After at most T deletions the retained original blocks are linear. A deletion loses at most g_D weight, so retained weight satisfies

$$F' \geq \binom{q}{2} - qr + n - (g_D - 1)T. \quad (10)$$

Only auxiliary blocks are deleted: original edges, their cover number and the counts in (8) are not changed by this operation.

Replicate a retained size- d block $d-1$ times and give each copy weight $(d-2)/2$. Its total copy weight is exactly g_d . Since the original blocks are linear, the degree of a point in the replicated system is $\sum_{B \ni p} (|B| - 1) \leq q - 1$. Its pair codegree is at most $D - 1$. Its rank is at most D . Also

$$0 \leq F' \leq \sum_B \binom{|B|}{2} \leq \binom{q}{2}. \quad (11)$$

For $q \geq q_0$, Input K colours these copies with at most $(1+\eta)(q-1)$ colours. If $F' = 0$, take the empty matching; otherwise some class has weight at least $F'/((1+\eta)(q-1))$. That class is a matching of original blocks and never uses two copies of the same block. Select the corresponding original vertices and pair all remaining original edges. Each pair has an intersection vertex. The matching saves $(d-2)/2$ per size- d block over the pairing baseline; the rounding cost is at most $1/2$. From (10)–(11) it follows that

$$\tau(J) \leq \frac{qr - n}{q - 1} + (g_D - 1) \frac{T}{q - 1} + \frac{\eta q}{2} + \frac{1}{2}. \quad (12)$$

Indeed replacing the factor $1/(1+\eta)$ by 1 costs at most $\eta F'/(q-1) \leq \eta q/2$. This reasoning is valid also when the lower bound in (10) is negative.

For every positive integer k , the integer inequality $(d-k)(d-k-1) \geq 0$ gives

$$n \geq \frac{2kqr - q(q-1) - 2T}{k(k+1)}. \quad (13)$$

If $q > r$, insert (13) into (12). The main terms are

$$r \left(\frac{k-1}{k+1} + \frac{q/r}{k(k+1)} \right),$$

plus $r(k-1)/((k+1)(q-1)) \leq 1$. The excess coefficient is $g_D - 1 + 2/(k(k+1)) \leq g_D$, and $q-1 \geq r$. Minimizing over k gives

$$\tau(J) \leq rh(q/r) + g_D \frac{T}{r} + \frac{\eta q}{2} + B_0, \quad B_0 = q_0/2 + 2. \quad (14)$$

For $q \leq r$, pairing gives $\tau(J) \leq q/2 + 1/2 = rh(q/r) + 1/2$. For $q < q_0$, pairing gives $\tau(J) \leq q_0/2$. These cases justify (14) for *all* $q \geq 0$; no colouring theorem is applied below its threshold.

For later use Cauchy–Schwarz applied to (8) gives a second lower bound:

$$n \geq \frac{q^2 r^2}{q(q+r-1) + 2T}, \quad q > 0. \quad (15)$$

4. Peeling and the envelope inequality

Select a vertex of current degree at least $L = D + 1$, delete its current incident edges, and continue until the residual J has maximum degree at most D . Let b vertices have been selected and q edges remain. Then

$$m \geq q + Lb, \quad \tau(H) \leq b + \tau(J), \quad I(J) \leq I(H). \quad (16)$$

Deleting edges preserves intersecting uniformity; an empty residual is covered by the conventions above. Since $q \leq m \leq Cr$, every slope of h on $[0, C]$ is at least $1/(K(K+1)) = 1/L$. Thus

$$\frac{m-q}{L} + rh(q/r) \leq rh(m/r). \quad (17)$$

Equivalently, restrict the minimum in (2) to $1 \leq k \leq K$: each of these affine functions has slope at least $1/L$, so their minimum has the same lower slope bound. Now combine (14), (16), (17), and choose $\eta = 2\varepsilon/C$. This proves (4) with $B_{C,\varepsilon} = q_0(D, 2\varepsilon/C)/2 + 2$.

For clarity the breakpoints in (2) follow from

$$h_k(x) - h_{k+1}(x) = \frac{2(x-k)}{k(k+1)(k+2)}.$$

Comparison with the harmonic curve is also exact:

$$\frac{x}{1+x} - h_k(x) = \frac{(x-k+1)(k-x)}{k(k+1)(1+x)}, \quad k-1 \leq x \leq k. \quad (18)$$

Corollary 2 follows by dividing (4) by r , using boundedness of the edge/rank ratio, and then letting $\varepsilon \downarrow 0$. If that ratio fails to tend to infinity while $\tau/r \rightarrow 1$, there is a bounded-ratio subsequence, contradicting (5).

5. Integer equality and degree concentration

We first record a peeling fact used twice. Suppose $m_j/r_j \rightarrow c$ and $\tau(H_j)/r_j \rightarrow h(c)$. Choose a fixed $C > c$ and a fixed D so large that $1/(D+1) < 1/(K(K+1))$, with $K = \max\{2, \lceil C \rceil\}$. Run the degree- $D+1$ peeling without depending on η . Apply (14) for this fixed D and arbitrary $\eta > 0$. The strict version of (17) is

$$rh(m/r) - \left(\frac{m-q}{D+1} + rh(q/r) \right) \geq \left(\frac{1}{K(K+1)} - \frac{1}{D+1} \right) (m-q). \quad (19)$$

Its coefficient is fixed and positive. Because $I = o(r^2)$, first take limits with D, η fixed, and then let $\eta \downarrow 0$. Equality forces $m-q = o(r)$, hence $b = o(r)$, $q/r \rightarrow c$, and $\tau(J)/r \rightarrow h(c)$. No uniformity of the colouring threshold in D or η is required.

Now take $c = a$ integer. Inequality (12), with arbitrarily small fixed η , implies

$$\limsup n(J)/r^2 \leq a - a \frac{a}{a+1} = \frac{a}{a+1}.$$

Inequality (15) gives the reverse lower limit. Removing $m - q = o(r)$ edges can remove at most $r(m - q) = o(r^2)$ active vertices, so the original active-vertex count has the same limit. Finally the exact identity on the original family is

$$\sum_v (d(v) - a - 1)^2 = m(m - 1) + mr + 2I - 2(a + 1)mr + (a + 1)^2 n. \quad (20)$$

Divide by r^2 and insert the established limits. Its limit is zero, proving (6).

6. Why equality fails between integer endpoints

Suppose instead $k - 1 < c < k$ with $k \geq 2$, and assume equality in (5). The peeling fact above and inequalities (12)–(13) imply

$$\frac{n(H)}{r^2} \longrightarrow \frac{c(2k - c)}{k(k + 1)}.$$

The lower limit uses (13), and the upper limit uses (12); passage from residual to original vertices again costs at most $r(m - q)$. Using (8) on the original family, we obtain

$$\sum_v (d(v) - k)(d(v) - k - 1) = o(r^2). \quad (21)$$

All summands are nonnegative because degrees are integers. Call a vertex bad if its degree is outside $\{k, k + 1\}$. For fixed k , both d^2 and $d|d - k|$, for bad positive integers d , are bounded by a fixed multiple of $(d - k)(d - k - 1)$. To see this, the ratio is bounded as $d \rightarrow \infty$, and the remaining bad integers form a finite set with strictly positive denominator. Therefore

$$\sum_{v \text{ mbad}} d(v)^2 = o(r^2), \quad \sum_{v \text{ mbad}} d(v)|d(v) - k| = o(r^2). \quad (22)$$

Use the original m edges as auxiliary points and let Q consist of the blocks of vertices of degree exactly $k + 1$. For an original edge A , write $i_A = \sum_{B \neq A} (|A \cap B| - 1)$, so $\sum_A i_A = 2I = o(r^2)$. Its degree in Q satisfies the exact identity

$$d_Q(A) = m - 1 - (k - 1)r + i_A - \sum_{v \in A, v \text{ mbad}} (d(v) - k). \quad (23)$$

Consequently, with $\theta = c - k + 1 \in (0, 1)$,

$$\sum_A |d_Q(A) - \theta r| = o(r^2). \quad (24)$$

This uses (22), $\sum i_A = 2I$, and $m/r \rightarrow c$; it is an L^1 estimate, not an assertion that every point is regular.

The pair excess of Q is at most $I(H)$. Delete at most $I(H) = o(r^2)$ blocks to make it linear, as in Section 3. This changes the total degree sum by at most $(k + 1)I = o(r^2)$, so (24) still holds for the resulting Q' . Each point has degree at most r , since its blocks correspond to vertices in one original r -edge. Fix $\xi > 0$. Remove every block meeting a point of degree greater than $(\theta + \xi)r$. By (24), there

are $o(r)$ such points, and this removes $o(r^2)$ blocks. The remaining linear $(k+1)$ -uniform auxiliary graph has

$$\Delta \leq (\theta + \xi)r, \quad |E| = \frac{m\theta r}{k+1} + o(r^2), \quad \Delta_2 \leq 1. \quad (25)$$

For fixed $\eta, \xi > 0$, Input K applies to (25) for all sufficiently large r . A largest colour class is a matching of size at least

$$\frac{m\theta}{(k+1)(1+\eta)(\theta+\xi)} + o(r).$$

Its defining original vertices, followed by pairing all remaining original edges, give a cover of size at most $m/2 - (k-1)|M|/2 + 1/2$. Let $r \rightarrow \infty$ first and then $\eta, \xi \downarrow 0$. It follows that

$$\limsup \tau(H)/r \leq \frac{c}{k+1} < h(c), \quad h(c) - \frac{c}{k+1} = \frac{(k-1)(k-c)}{k(k+1)} > 0. \quad (26)$$

This contradicts the assumed equality.

6.1 An explicit gap

For $k-1 < c < k$, $k \geq 2$, define

$$P = k(k+1), \quad \theta = c - k + 1, \quad \Delta = \frac{(k-1)(k-c)}{P}, \quad A = 2(k+2)P^2, \quad J = 1 + \frac{k + A(k+2)}{2\theta}. \quad (27)$$

We can take

$$\zeta_c = (\Delta/J)^2 > 0. \quad (28)$$

Here is a quantitative version of the previous argument. Pass to a subsequence on which $\tau(H)/r \rightarrow \alpha$ is its upper limit, and put $\delta = h(c) - \alpha$. Theorem 1 gives $\delta \geq 0$, and $\alpha \geq 0$ gives $\delta < 1$. Fix the peeling threshold $D+1 = 2P$. For all sufficiently large indices, $m/r < k$, and every slope of h up to m/r is at least $1/P$. The residual bound (14), followed by $\eta \downarrow 0$, therefore gives

$$\limsup (m - q)/r \leq 2P\delta. \quad (29)$$

We claim

$$\limsup n(H)/r^2 \leq c(1 - h(c)) + 2P\delta. \quad (30)$$

To verify it, pass further to a subsequence with $(m - q)/r \rightarrow z$, so $0 \leq z \leq 2P\delta$ and $q/r \rightarrow c - z$. Peeling gives $\liminf \tau(J)/r \geq \alpha - z/(2P)$. If $c - z > 0$, (12), again with fixed errors sent to zero after limits, gives

$$\limsup n(J)/r^2 \leq (c - z)(1 - \alpha + z/(2P)).$$

Adding at most $r(m - q)$ lost vertices bounds the original limit by $c - (c - z)(h(c) - \delta - z/(2P))$. This is at most

$$c(1 - h(c)) + (2c + 2Ph(c))\delta \leq c(1 - h(c)) + 2P\delta,$$

because $2c + 2Ph(c) = 2k(k-1) + 4c \leq 2P$. If $c - z = 0$, use $n(J) \leq qr = o(r^2)$; the same bound follows by the same expression with its zero factor. Subsequence selection proves (30).

The original integer degree polynomial now yields

$$\limsup \frac{1}{r^2} \sum_v (d(v) - k)(d(v) - k - 1) \leq 2P^2\delta. \quad (31)$$

For every bad positive integer degree,

$$d|d - k| \leq (k + 2)(d - k)(d - k - 1).$$

For $d \geq k + 2$, divide by $(d - k)(d - k - 1)$: the ratio is $d/(d - k - 1) \leq k + 2$. For $d \leq k - 1$ the ratio is $d/(k + 1 - d) \leq (k - 1)/2$. Thus (23) gives

$$\limsup \frac{1}{r^2} \sum_A |d_Q(A) - \theta r| \leq A\delta. \quad (32)$$

Deleting the $o(r^2)$ blocks needed for linearity changes (32) by $o(1)$ after normalization. For a fixed $\xi > 0$, the number of points with degree above $(\theta + \xi)r$, divided by r , has upper limit at most $A\delta/\xi$. Removing all their blocks costs at most $(A\delta/\xi + o(1))r^2$ blocks. The remaining linear $(k + 1)$ -uniform graph has maximum degree at most $(\theta + \xi)r$ and edge count with lower limit, divided by r^2 , at least

$$\frac{c\theta - A\delta}{k + 1} - \frac{A\delta}{\xi}.$$

Colouring and pairing as before, and then letting the fixed colouring error tend to zero, give

$$\alpha \leq \frac{c}{k + 1} + \frac{k - 1}{2(k + 1)} \frac{c\xi + A\delta(1 + (k + 1)/\xi)}{\theta + \xi}. \quad (33)$$

This inequality remains valid if the edge-count lower bound is negative: a matching of nonnegative size still satisfies that lower bound.

If $\delta = 0$, send $\xi \downarrow 0$ in (33) to contradict $h(c) > c/(k + 1)$. Otherwise choose $\xi = \sqrt{\delta}$. Since $\delta < 1$, $c < k$, and $(k - 1)/(2(k + 1)) \leq 1/2$, (33) implies

$$\Delta \leq \delta + \frac{k\sqrt{\delta} + A\delta + A(k + 1)\sqrt{\delta}}{2\theta} \leq J\sqrt{\delta}.$$

Therefore $\delta \geq (\Delta/J)^2$. This proves (7)–(28) for every subsequential upper limit, hence for the original sequence.

7. Explicit equality families

Let s be a prime power and $V = \mathbb{F}_s^N$, $N \rightarrow \infty$. For each one-dimensional direction, create a part consisting of all affine lines in that direction. A point of V defines the edge containing its line in every direction. Two distinct points lie on exactly one common affine line. Thus the resulting hypergraph is simple, intersecting, linear and partite, with

$$m = s^N, \quad r = \frac{s^N - 1}{s - 1}, \quad n = rs^{N-1}, \quad d(v) = s.$$

Every original vertex covers exactly s edges, while all lines in one parallel class cover the point set. Hence $\tau = s^{N-1}$, proving

$$m/r \rightarrow s - 1, \quad \tau/r \rightarrow (s - 1)/s, \quad n/r^2 \rightarrow (s - 1)/s.$$

These families attain Theorem 3's integer endpoint and degree profile.

For any $0 < c \leq 1$, choose $m/r \rightarrow c$ with $r \geq m - 1$. Start with m edges indexed by $1, \dots, m$. Give each pair i, j its own vertex incident only with edges i, j , and add private vertices to each edge until it has size r . The family is simple and linear, and every active vertex has degree at most two. Thus $\tau \geq \lceil m/2 \rceil$; pairing attains this value. It gives $\tau/r \rightarrow c/2 = h(c)$. The endpoint $c = 0$ follows, for example, with bounded m and increasing r . This last construction need not be partite; no such requirement is made here.

8. Verification scope and remaining questions

The universal proof is Sections 2–6. The code supplies exact incidence, block-deletion, replication and matching-cover diagnostics, including nonpartite examples. Lean checks selected universal scalar identities and inequalities, with the exact exported statements recorded in the source. It does not formalize Input K, finite hypergraph covers, the asymptotic arguments or the entire theorems. Hashes record the payload; hash consistency itself is not mathematical verification.

The unresolved questions include the best gap in (7), the true upper frontier between integer ratios, effective growth rates in the near-linear Erdős–Lovász edge minimum, and the general nonlinear partite minimum. The theorem supplies no bound when the excess is uncontrolled and supplies no counterexample to Ryser’s conjecture.

References

- [KKKMO] D. Y. Kang, T. Kelly, D. Kühn, A. Methuku and D. Osthus, *Solution to a problem of Erdős on the chromatic index of hypergraphs with bounded codegree*, Proceedings of the London Mathematical Society **129** (2024), no. 6, e70011. Primary manuscript <https://arxiv.org/abs/2110.06181>, Section 1.2 and Theorem 3.1 (Kahn’s small-codegree theorem).
- [K] J. Kahn, *Asymptotically good list-colorings*, Journal of Combinatorial Theory, Series A **73** (1996), 1–59.
- [S] V. Sivashankar, *An Improved Lower Bound for the Erdős–Lovász Cover Number Problem*, 2026 preprint, <https://arxiv.org/abs/2606.24878v2>, especially Section 4. The degree-peeling/linearization context is attributed to that source; its degree-three residual lemma is not used in this note.
- [ABW] R. Aharoni, J. Barát and I. M. Wanless, *Multipartite hypergraphs achieving equality in Ryser’s conjecture*, Graphs and Combinatorics **32** (2016), 1–15; <https://arxiv.org/abs/1409.4833>.
- [Previous] Repository note *Near-linear Ryser equality families require superlinear edge counts*, frozen at commit `ea1f13d`, and its cited predecessors. The present statements drop the partite restriction, strengthen the bounded-ratio cover gap, and address equality regimes.